



EXPRESS.JS: THE DEFINITIVE NODE.JS FRAMEWORK GUIDE

A comprehensive guide to its core components and flow.



EXPRESS OVERVIEW



MINIMALIST & FAST



UNOPINIONATED DESIGN



BUILT ON NODE.JS



GREAT FOR REST APIS

```
app.use((req, res, next) => {  
  ...  
  next();  
});
```

MIDDLEWARE PIPELINE



PROCESSING CORE

ROUTE MATCHING NEXUS



ROUTE HANDLER PODS

HOME HANDLER

```
app.get('/')  
HTML
```

USER PROFILE HANDLER

```
app.get('/users/:id')  
JSON :id
```

LOGIN HANDLER

```
app.post('/login')  
HTML : Data
```

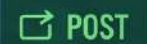
SERVER
PROCESSING
CORE

ROUTING ELEMENTS

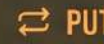
METHODS



GET



POST



PUT



DELETE

PATHS



/



/...



...

PARAMETERS

req.params

MIDDLEWARE ANATOMY

```
(req, res, next) {  
  con (res, next);  
  ... next();  
});
```

- Detailed code (function function, nex, res replament and respons functions and flow.
- Explanations of authentications, error handling and unciomusts.

PERFORMANCE & FEATURES



FAST
PERFORMANCE



EASY
EXTENSIBILITY



STRONG
ECOSYSTEM

ROUTING PATTERN ANATOMY

```
app.method(path, handler)
```

Example of
(path, handler)

Example of
parameters

Examples, name
(paths, parameters)



UNDERSTANDING NODE.JS ROUTING & MIDDLEWARE

ROUTING IN NODE.JS



HOW ROUTING WORKS

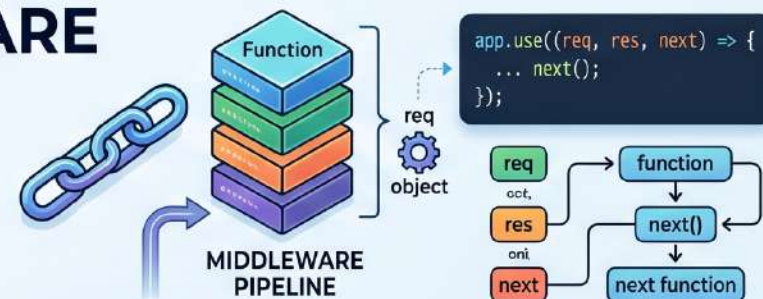
Incoming e.g. `GET /api/v1/products` gets matched.



URL PARAMETERS



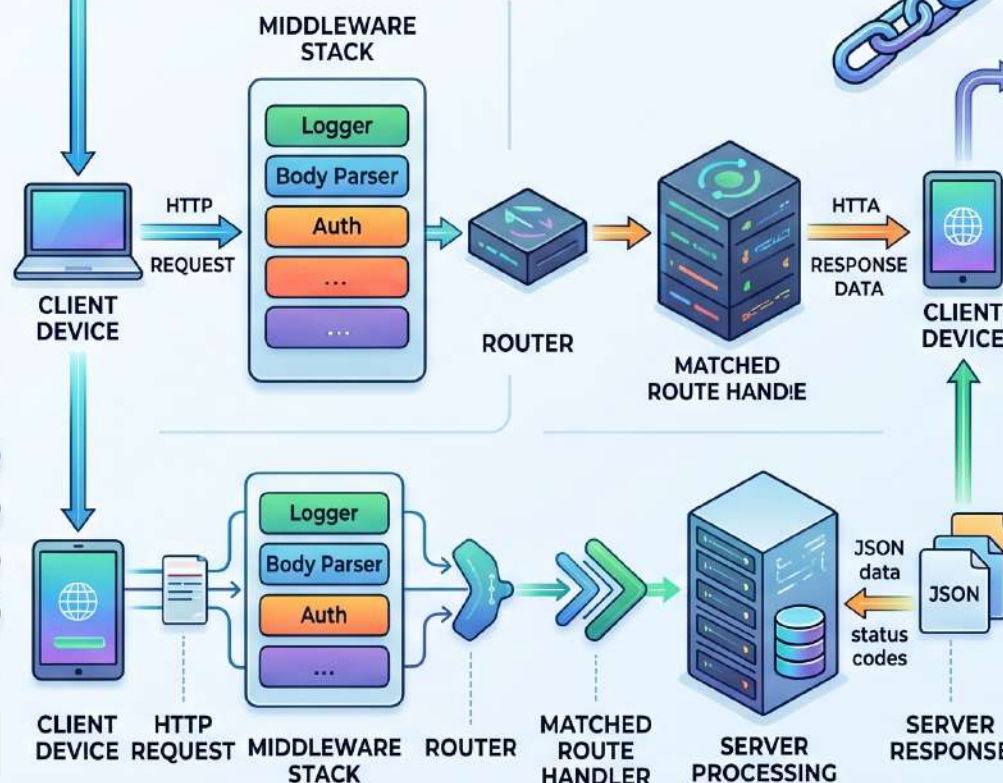
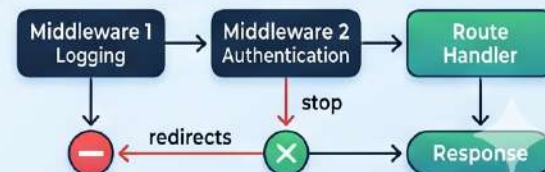
MIDDLEWARE IN NODE.JS



TYPES OF MIDDLEWARE



ORDER MATTERS



01_app.js U X JS 02_res-send.js U JS 03_middleware_routing. ▶ 🔍 📄 ...

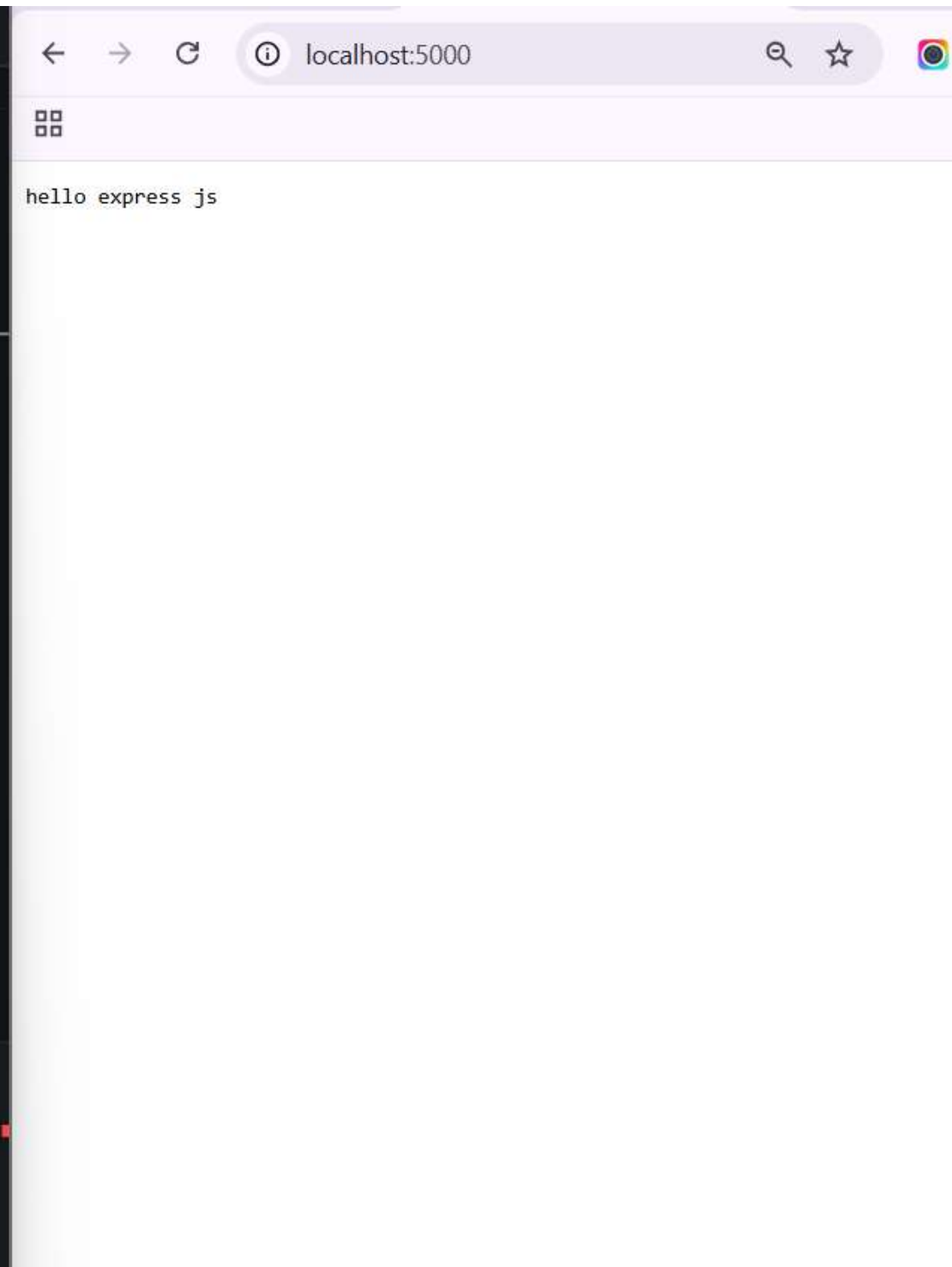
03_express > JS 01_app.js > ...

```
1
2 import express from 'express';
3
4 // const express = require('express'); ->commonjs
5
6 const app = express();
7
8 app.get("/",(req,res)=>{
9
10     res.end("hello express js");
11 });
12
13 const port = 5000;
14
15 app.listen(port,(err)=>{
16     if(err){
17         return console.log(err.message);
18     }
19     console.log(`server running on port ${port}`);
20 })
```

PROBLEMS TERMINAL ...

node - 03_express + - 📄 🗑️ ... | 🔍 ✕

PS C:\Nodejs-backend\03_express> node 01_app.js
server running on port 5000
█



```

1
2 import express from 'express';
3
4 const app = express();
5
6 app.get("/",(req,res)=>{
7   |   res.send("welcome to express js");
8   | });
9
10 const port = 5000;
11
12 app.listen(port,(err)=>{
13   |   if(err){
14   |     |   return console.log(err.message);
15   |   }
16   |   console.log(`server running on port ${port}`);
17   | })

```

```
PS C:\Nodejs-backend\03_express> npm run dev
> 03_express@1.0.0 dev
> node --watch 02_res-send.js

server running on port 5000

```


03_express > JS 03_middleware_routing.js > ...

```
3 import express from 'express';
4
5 const app = express();
6
7 //Middleware (run before every request)
8
9 app.use((req,res,next)=>{
10   console.log(`Request URL: ${req.url}`);
11   next();
12 });
13
14 //Routing
15
16 app.get("/",(req,res)=>{
17   res.send("Home page");
18 });
19
20 app.get("/about",(req,res)=>{
21   res.send("About page");
22 });
23
24 app.get("/contact",(req,res)=>{
25   res.send("contact page");
26 });
27
28 //server start
29
30 const port = 5000;
31
32 app.listen(port,(err)=>{
33   if(err){
```

PROBLEMS

TERMINAL

node - 03_express

```
PS C:\Nodejs-backend\03_express> npm run dev
sercer running on port 5000
Request URL: /
```



localhost:5000



Home page



About page

03_express > JS 03_middleware_routing.js > ...

```
3 import express from 'express';
4
5 const app = express();
6
7 //Middleware (run before every request)
8
9 app.use((req,res,next)=>{
10   console.log(`Request URL: ${req.url}`);
11   next();
12 });
13
14 //Routing
15
16 app.get("/",(req,res)=>{
17   res.send("Home page");
18 });
19
20 app.get("/about",(req,res)=>{
21   res.send("About page");
22 });
23
24 app.get("/contact",(req,res)=>{
25   res.send("contact page");
26 });
27
28 //server start
29
30 const port = 5000;
31
32 app.listen(port,(err)=>{
33   if(err){
```

PROBLEMS TERMINAL ... node - 03_express + v 🗑 ... | 🔍 ✕

PS C:\Nodejs-backend\03_express> npm run dev

Request URL: /

03_express > JS 03_middlewares_routing.js > ...

```

3  import express from 'express';
4
5  const app = express();
6
7  //Middleware (run before every request)
8
9  app.use((req,res,next)=>{
10 |   console.log(`Request URL: ${req.url}`);
11 |   next();
12 | });
13
14  //Routing
15
16  app.get("/",(req,res)=>{
17 |   res.send("Home page");
18 | });
19
20  app.get("/about",(req,res)=>{
21 |   res.send("About page");
22 | });
23
24  app.get("/contact",(req,res)=>{
25 |   res.send("contact page");
26 | });
27
28  //server start
29
30  const port = 5000;
31
32  app.listen(port,(err)=>{
33 |   if(err){

```

contact page

PROBLEMS TERMINAL ... node - 03_express + v

PS C:\Nodejs-backend\03_express> npm run dev

Request URL: /about

Request URL: /contact